MATHHUB_PRO // NCERT_SOLUTIONS

ANSWERS

EXERCISE 1.1

1. (i) Neither reflexive nor symmetric nor transitive.

(ii) Neither reflexive nor symmetric but transitive.

(iii) Reflexive and transitive but not symmetric.

(iv) Reflexive, symmetric and transitive.

(v) (a) Reflexive, symmetric and transitive.

(b) Reflexive, symmetric and transitive.

(c) Neither reflexive nor symmetric nor transitive.

(d) Neither reflexive nor symmetric but transitive.

(e) Neither reflexive nor symmetric nor transitive.

3. Neither reflexive nor symmetric nor transitive.

5. Neither reflexive nor symmetric nor transitive.

9. (i) $\{1, 5, 9\}$, (ii) $\{1\}$

12. T_1 is related to T_3.

13. The set of all triangles

14. The set of all lines $y = 2x + c$, $c \in \mathbf{R}$

15. B

16. C

EXERCISE 1.2

1. No

2. (i) Injective but not surjective (ii) Neither injective nor surjective

(iii) Neither injective nor surjective (iv) Injective but not surjective

(v) Injective but not surjective

7. (i) One-one and onto (ii) Neither one-one nor onto.

9. No **10.** Yes **11.** D **12.** A

Miscellaneous Exercise on Chapter 1

3. No **4.** $n!$ **5.** Yes **6.** A **7.** B

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EXERCISE 2.1

1. $\frac{-\pi}{6}$ 2. $\frac{\pi}{6}$ 3. $\frac{\pi}{6}$ 4. $\frac{-\pi}{3}$

5. $\frac{2\pi}{3}$ 6. $-\frac{\pi}{4}$ 7. $\frac{\pi}{6}$ 8. $\frac{\pi}{6}$

9. $\frac{3\pi}{4}$ 10. $\frac{-\pi}{4}$ 11. $\frac{3\pi}{4}$ 12. $\frac{2\pi}{3}$

13. B 14. B

EXERCISE 2.2

3. $\frac{1}{2}\tan^{-1} x$ 4. $\frac{x}{2}$ 5. $\frac{\pi}{4} - x$

6. $\sin^{-1}\frac{x}{a}$ 7. $3\tan^{-1}\frac{x}{a}$ 8. $\frac{\pi}{4}$

9. $\frac{x+y}{1-xy}$ 10. $\frac{\pi}{3}$

11. $\frac{-\pi}{4}$ 12. $\frac{17}{6}$ 13. B 14. D

15. B

Miscellaneous Exercise on Chapter 2

1. $\frac{\pi}{6}$ 2. $\frac{\pi}{6}$ 11. $x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$ 12. $x = \frac{1}{\sqrt{3}}$

13. D 14. C

EXERCISE 3.1

1. (i) 3×4 (ii) 12 (iii) $19, 35, -5, 12, \frac{5}{2}$

2. $1 \times 24, 2 \times 12, 3 \times 8, 4 \times 6, 6 \times 4, 8 \times 3, 12 \times 2, 24 \times 1; 1 \times 13, 13 \times 1$

3. $1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 18 \times 1; 1 \times 5, 5 \times 1$

4. (i) $\begin{bmatrix} 2 & \frac{9}{2} \\ \frac{9}{2} & 8 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & \frac{1}{2} \\ 2 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} \frac{9}{2} & \frac{25}{2} \\ 8 & 18 \end{bmatrix}$

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5. (i) $\begin{bmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{5}{2} & 2 & \frac{3}{2} & 1 \\ 4 & \frac{7}{2} & 3 & \frac{5}{2} \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 0 & -1 & -2 \\ 3 & 2 & 1 & 0 \\ 5 & 4 & 3 & 2 \end{bmatrix}$

6. (i) $x = 1, \quad y = 4, \quad z = 3$

(ii) $x = 4, \quad y = 2, \quad z = 0$ or $x = 2, \quad y = 4, \quad z = 0$

(iii) $x = 2, \quad y = 4, \quad z = 3$

7. $a = 1, b = 2, c = 3, d = 4$

8. C **9.** B **10.** D

EXERCISE 3.2

1. (i) $A + B = \begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix}$ (ii) $A - B = \begin{bmatrix} 1 & 1 \\ 5 & -3 \end{bmatrix}$

(iii) $3A - C = \begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix}$ (iv) $AB = \begin{bmatrix} -6 & 26 \\ -1 & 19 \end{bmatrix}$ (v) $BA = \begin{bmatrix} 11 & 10 \\ 11 & 2 \end{bmatrix}$

2. (i) $\begin{bmatrix} 2a & 2b \\ 0 & 2a \end{bmatrix}$ (ii) $\begin{bmatrix} (a+b)^2 & (b+c)^2 \\ (a-c)^2 & (a-b)^2 \end{bmatrix}$

(iii) $\begin{bmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

3. (i) $\begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{bmatrix}$ (iii) $\begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}$

(iv) $\begin{bmatrix} 14 & 0 & 42 \\ 18 & -1 & 56 \\ 22 & -2 & 70 \end{bmatrix}$ (v) $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 5 \\ -2 & 2 & 0 \end{bmatrix}$ (vi) $\begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix}$

4. $A + B = \begin{bmatrix} 4 & 1 & -1 \\ 9 & 2 & 7 \\ 3 & -1 & 4 \end{bmatrix}, B - C = \begin{bmatrix} -1 & -2 & 0 \\ 4 & -1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$

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5. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ **6.** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

7. (i) $X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}, Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ (ii) $X = \begin{bmatrix} \frac{2}{5} & \frac{-12}{5} \\ \frac{-11}{5} & 3 \end{bmatrix}, Y = \begin{bmatrix} \frac{2}{5} & \frac{13}{5} \\ \frac{14}{5} & -2 \end{bmatrix}$

8. $X = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$ **9.** $x = 3, y = 3$ **10.** $x = 3, y = 6, z = 9, t = 6$

11. $x = 3, y = -4$ **12.** $x = 2, y = 4, w = 3, z = 1$

15. $\begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$ **17.** $k = 1$

19. (a) ₹15000, ₹15000 (b) ₹5000, ₹25000

20. ₹20160 **21.** A **22.** B

EXERCISE 3.3

1. (i) $\begin{bmatrix} 5 & \frac{1}{2} & -1 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ (iii) $\begin{bmatrix} -1 & \sqrt{3} & 2 \\ 5 & 5 & 3 \\ 6 & 6 & -1 \end{bmatrix}$

4. $\begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix}$ **9.** $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$

10. (i) $A = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

(ii) $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

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(iii) $A = \begin{bmatrix} 3 & \frac{1}{2} & \frac{-5}{2} \\ \frac{1}{2} & -2 & -2 \\ \frac{-5}{2} & -2 & 2 \end{bmatrix} + \begin{bmatrix} 0 & \frac{5}{2} & \frac{3}{2} \\ \frac{-5}{2} & 0 & 3 \\ \frac{-3}{2} & -3 & 0 \end{bmatrix}$ (iv) $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$

11. A **12.** B

EXERCISE 3.4

1. D

Miscellaneous Exercise on Chapter 3

3. $x = \pm \frac{1}{\sqrt{2}}, y = \pm \frac{1}{\sqrt{6}}, z = \pm \frac{1}{\sqrt{3}}$

4. $x = -1$ **6.** $x = \pm 4\sqrt{3}$

7. (a) Total revenue in the market - I = ₹46000

Total revenue in the market - II = ₹53000

(b) ₹15000, ₹17000

8. $X = \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$ **9.** C **10.** B **11.** C

EXERCISE 4.1

1. (i) 18 **2.** (i) 1, (ii) $x^3 - x^2 + 2$

5. (i) – 12, (ii) 46, (iii) 0, (iv) 5 **6.** 0

7. (i) $x = \pm\sqrt{3}$, (ii) $x = 2$ **8.** (B)

EXERCISE 4.2

1. (i) $\frac{15}{2}$, (ii) $\frac{47}{2}$, (iii) 15

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3. (i) 0, 8, (ii) 0, 8 **4.** (i) $y = 2x$, (ii) $x - 3y = 0$ **5.** (D)

EXERCISE 4.3

1. (i) $M_{11} = 3, M_{12} = 0, \quad M_{21} = -4, M_{22} = 2, A_{11} = 3, A_{12} = 0, A_{21} = 4, A_{22} = 2$

(ii) $M_{11} = d, M_{12} = b, \quad M_{21} = c, \quad M_{22} = a$

$A_{11} = d, \quad A_{12} = -b, A_{21} = -c, \quad A_{22} = a$

2. (i) $M_{11} = 1, M_{12} = 0, M_{13} = 0, \quad M_{21} = 0, M_{22} = 1, M_{23} = 0, M_{31} = 0, M_{32} = 0, M_{33} = 1,$

$A_{11} = 1, A_{12} = 0, \quad A_{13} = 0, A_{21} = 0, A_{22} = 1, A_{23} = 0, \quad A_{31} = 0, \quad A_{32} = 0, \quad A_{33} = 1$

(ii) $M_{11} = 11, M_{12} = 6, \quad M_{13} = 3, M_{21} = -4, M_{22} = 2, M_{23} = 1, \quad M_{31} = -20, M_{32} = -13, M_{33} = 5$

$A_{11} = 11, \quad A_{12} = -6, A_{13} = 3, \quad A_{21} = 4, \quad A_{22} = 2, \quad A_{23} = -1, A_{31} = -20, A_{32} = 13, A_{33} = 5$

3. 7 **4.** $(x - y)(y - z)(z - x)$ **5.** (D)

EXERCISE 4.4

1. $\begin{matrix} 4 & -2 \\ -3 & 1 \end{matrix}$ **2.** $\begin{matrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{matrix}$ **5.** $\frac{1}{14}\begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$

6. $\frac{1}{13}\begin{matrix} 2 & -5 \\ 3 & -1 \end{matrix}$ **7.** $\frac{1}{10}\begin{bmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{bmatrix}$ **8.** $\frac{-1}{3}\begin{matrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{matrix}$

9. $\frac{-1}{3}\begin{bmatrix} -1 & 5 & 3 \\ -4 & 23 & 12 \\ 1 & -11 & -6 \end{bmatrix}$ **10.** $\begin{matrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{matrix}$ **11.** $\begin{matrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & \sin\alpha & -\cos\alpha \end{matrix}$

13. $\frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$ **14.** $a = -4, b = 1$ **15.** $A^{-1} = \frac{1}{11}\begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$

16. $\frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ **17.** B **18.** B

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EXERCISE 4.5

1. Consistent **2.** Consistent **3.** Inconsistent

4. Consistent **5.** Inconsistent **6.** Consistent

7. $x = 2, y = -3$ **8.** $x = \frac{-5}{11}, y = \frac{12}{11}$ **9.** $x = \frac{-6}{11}, y = \frac{-19}{11}$

10. $x = -1, y = 4$ **11.** $x = 1, y = \frac{1}{2}, z = \frac{-3}{2}$

12. $x = 2, y = -1, z = 1$ **13.** $x = 1, y = 2, z = -1$

14. $x = 2, y = 1, z = 3$

15. $\begin{matrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{matrix}$, $x = 1, y = 2, z = 3$

16. cost of onions per kg = ₹ 5
cost of wheat per kg = ₹ 8
cost of rice per kg = ₹ 8

Miscellaneous Exercise on Chapter 4

2. 1 **3.** $\begin{bmatrix} 9 & -3 & 5 \\ -2 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

5. $-2(x^3 + y^3)$ **6.** xy **7.** $x = 2, y = 3, z = 5$

8. A **9.** D

EXERCISE 5.1

2. f is continuous at $x = 3$

3. (a), (b), (c) and (d) are all continuous functions

5. f is continuous at $x = 0$ and $x = 2$; Not continuous at $x = 1$

6. Discontinuous at $x = 2$ **7.** Discontinuous at $x = 3$

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8. Discontinuous at $x = 0$

9. No point of discontinuity

10. No point of discontinuity

11. No point of discontinuity

12. f is discontinuous at $x = 1$

13. f is not continuous at $x = 1$

14. f is not continuous at $x = 1$ and $x = 3$

15. $x = 1$ is the only point of discontinuity

16. Continuous

17. $a = b + \frac{2}{3}$

18. For no value of λ, f is continuous at $x = 0$ but f is continuous at $x = 1$ for any value of λ.

20. f is continuous at $x = \pi$

21. (a), (b) and (c) are all continuous

22. Cosine function is continuous for all $x \in \mathbf{R}$; cosecant is continuous except for $x = n\pi$, $n \in \mathbf{Z}$; secant is continuous except for $x = (2n+1)\frac{\pi}{2}$, $n \in \mathbf{Z}$ and cotangent function is continuous except for $x = n\pi$, $n \in \mathbf{Z}$

23. There is no point of discontinuity.

24. Yes, f is continuous for all $x \in \mathbf{R}$

25. f is continuous for all $x \in \mathbf{R}$

26. $k = 6$

27. $k = \frac{3}{4}$

28. $k = \frac{-2}{\pi}$

29. $k = \frac{9}{5}$

30. $a = 2, b = 1$

34. There is no point of discontinuity.

EXERCISE 5.2

1. $2x \cos(x^2 + 5)$

2. $-\cos x \sin(\sin x)$

3. $a \cos(ax + b)$

4. $\dfrac{\sec(\tan\sqrt{x}).\tan(\tan\sqrt{x}).\sec^2\sqrt{x}}{2\sqrt{x}}$

5. $a \cos(ax + b) \sec(cx + d) + c \sin(ax + b) \tan(cx + d) \sec(cx + d)$

6. $10x^4 \sin x^5 \cos x^5 \cos x^3 - 3x^2 \sin x^3 \sin^2 x^5$

7. $\dfrac{-2\sqrt{2}\,x}{\sin x^2 \sqrt{\sin 2x^2}}$

8. $-\dfrac{\sin\sqrt{x}}{2\sqrt{x}}$

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EXERCISE 5.3

1. $\dfrac{\cos x - 2}{3}$ 2. $\dfrac{2}{\cos y - 3}$ 3. $-\dfrac{a}{2by + \sin y}$

4. $\dfrac{\sec^2 x - y}{x + 2y - 1}$ 5. $-\dfrac{(2x+y)}{(x+2y)}$ 6. $-\dfrac{(3x^2+2xy+y^2)}{(x^2+2xy+3y^2)}$

7. $\dfrac{y \sin xy}{\sin 2y - x \sin xy}$ 8. $\dfrac{\sin 2x}{\sin 2y}$ 9. $\dfrac{2}{1+x^2}$ 10. $\dfrac{3}{1+x^2}$

11. $\dfrac{2}{1+x^2}$ 12. $\dfrac{-2}{1+x^2}$ 13. $\dfrac{-2}{1+x^2}$ 14. $\dfrac{2}{\sqrt{1-x^2}}$

15. $-\dfrac{2}{\sqrt{1-x^2}}$

EXERCISE 5.4

1. $\dfrac{e^x(\sin x - \cos x)}{\sin^2 x}$, $x \neq n\pi$, $n \in \mathbf{Z}$ 2. $\dfrac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}, x \in (-1,1)$

3. $3x^2 e^{x^3}$ 4. $-\dfrac{e^{-x}\cos(\tan^{-1} e^{-x})}{1+e^{-2x}}$

5. $-e^x \tan e^x$, $e^x \neq (2n+1)\dfrac{\pi}{2}, n \in \mathbf{N}$ 6. $e^x + 2x^{e^{x^2}} + 3x^2 e^{x^3} + 4x^3 e^{x^4} + 5x^4 e^{x^5}$

7. $\dfrac{e^{\sqrt{x}}}{4\sqrt{x e^{\sqrt{x}}}}, x > 0$ 8. $\dfrac{1}{x \log x}, x > 1$

9. $-\dfrac{(x \sin x \cdot \log x + \cos x)}{x(\log x)^2}, x > 0$ 10. $-\dfrac{1}{x} + e^x \sin(\log x + e^x), x > 0$

EXERCISE 5.5

1. $-\cos x \cos 2x \cos 3x\,[\tan x + 2\tan 2x + 3\tan 3x]$

2. $\dfrac{1}{2}\sqrt{\dfrac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}\left[\dfrac{1}{x-1} + \dfrac{1}{x-2} - \dfrac{1}{x-3} - \dfrac{1}{x-4} - \dfrac{1}{x-5}\right]$

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3. $(\log x)^{\cos x}\left[\dfrac{\cos x}{x\log x}-\sin x\log(\log x)\right]$

4. $x^x(1+\log x)-2^{\sin x}\cos x\log 2$

5. $(x+3)(x+4)^2(x+5)^3(9x^2+70x+133)$

6. $\left(x+\dfrac{1}{x}\right)^x\left[\dfrac{x^2-1}{x^2+1}+\log\left(x+\dfrac{1}{x}\right)\right]+x^{1+\frac{1}{x}}\left(\dfrac{x+1-\log x}{x^2}\right)$

7. $(\log x)^{x-1}[1+\log x\,.\,\log(\log x)]+2x^{\log x-1}\,.\,\log x$

8. $(\sin x)^x(x\cot x+\log\sin x)+\dfrac{1}{2}\dfrac{1}{\sqrt{x-x^2}}$

9. $x^{\sin x}\left[\dfrac{\sin x}{x}+\cos x\log x\right]+(\sin x)^{\cos x}[\cos x\cot x-\sin x\log\sin x]$

10. $x^{x\cos x}[\cos x\,.\,(1+\log x)-x\sin x\log x]-\dfrac{4x}{(x^2-1)^2}$

11. $(x\cos x)^x[1-x\tan x+\log(x\cos x)]+(x\sin x)^{\frac{1}{x}}\left[\dfrac{x\cot x+1-\log(x\sin x)}{x^2}\right]$

12. $-\dfrac{yx^{y-1}+y^x\log y}{x^y\log x+xy^{x-1}}$

13. $\dfrac{y}{x}\left(\dfrac{y-x\log y}{x-y\log x}\right)$

14. $\dfrac{y\tan x+\log\cos y}{x\tan y+\log\cos x}$

15. $\dfrac{y(x-1)}{x(y+1)}$

16. $(1+x)(1+x^2)(1+x^4)(1+x^8)\left[\dfrac{1}{1+x}+\dfrac{2x}{1+x^2}+\dfrac{4x^3}{1+x^4}+\dfrac{8x^7}{1+x^8}\right]$; $f'(1)=120$

17. $5x^4-20x^3+45x^2-52x+11$

EXERCISE 5.6

1. t^2

2. $\dfrac{b}{a}$

3. $-4\sin t$

4. $-\dfrac{1}{t^2}$

5. $\dfrac{\cos\theta-2\cos 2\theta}{2\sin 2\theta-\sin\theta}$

6. $-\cot\dfrac{\theta}{2}$

7. $-\cot 3t$

8. $\tan t$

9. $\dfrac{b}{a}\operatorname{cosec}\theta$

10. $\tan\theta$

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EXERCISE 5.7

1. 2 **2.** $380\ x^{18}$ **3.** $-x \cos x - 2 \sin x$

4. $-\dfrac{1}{x^2}$ **5.** $x(5 + 6 \log x)$ **6.** $2e^x(5 \cos 5x - 12 \sin 5x)$

7. $9\ e^{6x}\ (3 \cos 3x - 4 \sin 3x)$ **8.** $-\dfrac{2x}{(1+x^2)^2}$

9. $-\dfrac{(1+\log x)}{(x \log x)^2}$ **10.** $-\dfrac{\sin(\log x) + \cos(\log x)}{x^2}$

12. $-\cot y \operatorname{cosec}^2 y$

Miscellaneous Exercise on Chapter 5

1. $27\ (3x^2 - 9x + 5)^8\ (2x - 3)$ **2.** $3\sin x \cos x\ (\sin x - 2 \cos^4 x)$

3. $(5x)^{3\cos 2x}\left[\dfrac{3\cos 2x}{x} - 6\sin 2x \log 5x\right]$

4. $\dfrac{3}{2}\sqrt{\dfrac{x}{1-x^3}}$ **5.** $-\left[\dfrac{1}{\sqrt{4-x^2}\sqrt{2x+7}} + \dfrac{\cos^{-1}\dfrac{x}{2}}{(2x+7)^{\frac{3}{2}}}\right]$

6. $\dfrac{1}{2}$ **7.** $(\log x)^{\log x}\left[\dfrac{1}{x} + \dfrac{\log(\log x)}{x}\right],\ x > 1$

8. $(a \sin x - b \cos x) \sin (a \cos x + b \sin x)$

9. $(\sin x - \cos x)^{\sin x - \cos x}\ (\cos x + \sin x)\ (1 + \log(\sin x - \cos x)),\ \sin x > \cos x$

10. $x^x\ (1 + \log x) + ax^{a-1} + a^x \log a$

11. $x^{x^2-3}\left[\dfrac{x^2-3}{x} + 2x\log x\right] + (x-3)^{x^2}\left[\dfrac{x^2}{x-3} + 2x\log(x-3)\right]$

12. $\dfrac{6}{5}\cot\dfrac{t}{2}$ **13.** 0 **17.** $\dfrac{\sec^3 t}{at}, 0 < t < \dfrac{\pi}{2}$

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EXERCISE 6.1

1. (a) 6π cm²/cm (b) 8π cm²/cm

2. $\frac{8}{3}$ cm²/s **3.** 60π cm²/s **4.** 900 cm³/s

5. 80π cm²/s **6.** 1.4π cm/s

7. (a) –2 cm/min (b) 2 cm²/min

8. $\frac{1}{\pi}$ cm/s **9.** 400π cm³/cm **10.** $\frac{8}{3}$ cm/s

11. (4, 11) and $\left(-4, \frac{-31}{3}\right)$ **12.** 2π cm³/s

13. $\frac{27}{8}\pi(2x+1)^2$ **14.** $\frac{1}{48\pi}$ cm/s **15.** ₹20.967

16. ₹208 **17.** B **18.** D

EXERCISE 6.2

4. (a) $\left(\frac{3}{4}, \infty\right)$ (b) $\left(-\infty, \frac{3}{4}\right)$

5. (a) $(-\infty, -2)$ and $(3, \infty)$ (b) $(-2, 3)$

6. (a) decreasing for $x < -1$ and increasing for $x > -1$

(b) decreasing for $x > -\frac{3}{2}$ and increasing for $x < -\frac{3}{2}$

(c) increasing for $-2 < x < -1$ and decreasing for $x < -2$ and $x > -1$

(d) increasing for $x < -\frac{9}{2}$ and decreasing for $x > -\frac{9}{2}$

(e) increasing in (1, 3) and $(3, \infty)$, decreasing in $(-\infty, -1)$ and $(-1, 1)$.

8. $0 < x < 1$ and $x > 2$ **12.** A, B

13. D **14.** $a > -2$ **19.** D

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EXERCISE 6.3

1. (i) Minimum Value = 3 (ii) Minimum Value = – 2

(iii) Maximum Value = 10 (iv) Neither minimum nor maximum value

2. (i) Minimum Value = – 1; No maximum value

(ii) Maximum Value = 3; No minimum value

(iii) Minimum Value = 4; Maximum Value = 6

(iv) Minimum Value = 2; Maximum Value = 4

(v) Neither minimum nor Maximum Value

3. (i) local minimum at $x = 0$, local minimum value = 0

(ii) local minimum at $x = 1$, local minimum value = – 2

local maximum at $x = -1$, local maximum value = 2

(iii) local maximum at $x = \frac{\pi}{4}$, local maximum value = $\sqrt{2}$

(iv) local maximum at $x = \frac{3\pi}{4}$, local maximum value = $\sqrt{2}$

local minimum at $x = \frac{7\pi}{4}$, local minimum value = $-\sqrt{2}$

(v) local maximum at $x = 1$, local maximum value = 19

local minimum at $x = 3$, local minimum value = 15

(vi) local minimum at $x = 2$, local minimum value = 2

(vii) local maximum at $x = 0$, local maximum value = $\frac{1}{2}$

(viii) local maximum at $x = \frac{2}{3}$, local maximum value = $\frac{2\sqrt{3}}{9}$

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5. (i) Absolute minimum value = – 8, absolute maximum value = 8

(ii) Absolute minimum value = – 1, absolute maximum value = $\sqrt{2}$

(iii) Absolute minimum value = – 10, absolute maximum value = 8

(iv) Absolute minimum value = 19, absolute maximum value = 3

6. Maximum profit = 113 unit.

7. Minima at $x = 2$, minimum value = – 39, Maxima at $x = 0$, maximum value = 25.

8. At $x = \frac{\pi}{4}$ and $\frac{5\pi}{4}$

9. Maximum value = $\sqrt{2}$

10. Maximum at $x = 3$, maximum value 89; maximum at $x = -2$, maximum value = 139

11. $a = 120$

12. Maximum at $x = 2\pi$, maximum value = 2π; Minimum at $x = 0$, minimum value = 0

13. 12, 12

14. 45, 15

15. 25, 10

16. 8, 8

17. 3 cm

18. $x = 5$ cm

21. radius = $\left(\frac{50}{\pi}\right)^{\frac{1}{3}}$ cm and height = $2\left(\frac{50}{\pi}\right)^{\frac{1}{3}}$ cm

22. $\frac{112}{\pi + 4}$ cm, $\frac{28\pi}{\pi + 4}$ cm

27. A

28. D

29. C

Miscellaneous Exercise on Chapter 6

2. $b\sqrt{3}$ cm²/s

3. (i) $0 \le x \le \frac{\pi}{2}$ and $\frac{3\pi}{2} < x < 2\pi$ (ii) $\frac{\pi}{2} < x < \frac{3\pi}{2}$

4. (i) $x < -1$ and $x > 1$ (ii) $-1 < x < 1$

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5. $\frac{3\sqrt{3}}{4}ab$ **6.** Rs 1000

8. length = $\frac{20}{\pi+4}$ m, breadth = $\frac{10}{\pi+4}$ m

10. (i) local maxima at $x=\frac{2}{7}$ (ii) local minima at $x=2$

(iii) point of inflection at $x=-1$

11. Absolute maximum = $\frac{5}{4}$, Absolute minimum = 1

14. $\frac{4\pi R^3}{3\sqrt{3}}$ **16.** A

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SUPPLEMENTARY MATERIAL

CHAPTER 5

Theorem 5 (To be on page 129 under the heading Theorem 5)

(i) Derivative of Exponential Function $f(x) = e^x$.

If $f(x) = e^x$, then

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{e^{x+\Delta x} - e^x}{\Delta x}$$

$$= e^x \cdot \lim_{\Delta x \to 0} \frac{e^{\Delta x} - 1}{\Delta x}$$

$$= e^x \cdot 1 \quad \left[\text{since } \lim_{h \to 0} \frac{e^h - 1}{h} = 1\right]$$

Thus, $\dfrac{d}{dx}(e^x) = e^x$.

(ii) Derivative of logarithmic function $f(x) = \log_e x$.

If $f(x) = \log_e x$, then

$$f'(x) = \lim_{\Delta x \to 0} \frac{\log_e(x + \Delta x) - \log_e x}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\log_e\left(1 + \dfrac{\Delta x}{x}\right)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{1}{x} \frac{\log_e\left(1 + \dfrac{\Delta x}{x}\right)}{\dfrac{\Delta x}{x}}$$

$$= \frac{1}{x} \quad \left[\text{since } \lim_{h \to 0} \frac{\log_e(1+h)}{h} = 1\right]$$

Thus, $\dfrac{d}{dx}\log_e x = \dfrac{1}{x}$.

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NOTES

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ANSWERS

EXERCISE 7.1

1. $-\frac{1}{2}\cos 2x$
2. $\frac{1}{3}\sin 3x$
3. $\frac{1}{2}e^{2x}$
4. $\frac{1}{3a}(ax+b)^3$
5. $-\frac{1}{2}\cos 2x-\frac{4}{3}e^{3x}$
6. $\frac{4}{3}e^{3x}+x+C$
7. $\frac{x^3}{3}-x+C$
8. $\frac{ax^3}{3}+\frac{bx^2}{2}+cx+C$
9. $\frac{2}{3}x^3+e^x+C$
10. $\frac{x^2}{2}+\log|x|-2x+C$
11. $\frac{x^2}{2}+5x+\frac{4}{x}+C$
12. $\frac{2}{7}x^{\frac{7}{2}}+2x^{\frac{3}{2}}+8\sqrt{x}+C$
13. $\frac{x^3}{3}+x+C$
14. $\frac{2}{3}x^{\frac{3}{2}}-\frac{2}{5}x^{\frac{5}{2}}+C$
15. $\frac{6}{7}x^{\frac{7}{2}}+\frac{4}{5}x^{\frac{5}{2}}+2x^{\frac{3}{2}}+C$
16. $x^2-3\sin x+e^x+C$
17. $\frac{2}{3}x^3+3\cos x+\frac{10}{3}x^{\frac{3}{2}}+C$
18. $\tan x + \sec x + C$
19. $\tan x - x + C$
20. $2\tan x - 3\sec x + C$
21. C
22. A

EXERCISE 7.2

1. $\log(1+x^2)+C$
2. $\frac{1}{3}(\log|x|)^3+C$
3. $\log|1+\log x|+C$
4. $\cos(\cos x)+C$
5. $-\frac{1}{4a}\cos 2(ax+b)+C$
6. $\frac{2}{3a}(ax+b)^{\frac{3}{2}}+C$
7. $\frac{2}{5}(x+2)^{\frac{5}{2}}-\frac{4}{3}(x+2)^{\frac{3}{2}}+C$

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8. $\frac{1}{6}(1+2x^2)^{\frac{3}{2}}+C$ **9.** $\frac{4}{3}(x^2+x+1)^{\frac{3}{2}}+C$ **10.** $2\log\left|\sqrt{x}-1\right|+C$

11. $\frac{2}{3}\sqrt{x+4}(x-8)+C$

12. $\frac{1}{7}(x^3-1)^{\frac{7}{3}}+\frac{1}{4}(x^3-1)^{\frac{4}{3}}+C$ **13.** $-\frac{1}{18(2+3x^3)^2}+C$

14. $\frac{(\log x)^{1-m}}{1-m}+C$ **15.** $-\frac{1}{8}\log|9-4x^2|+C$ **16.** $\frac{1}{2}e^{2x+3}+C$

17. $-\frac{1}{2e^{x^2}}+C$ **18.** $e^{\tan^{-1}x}+C$ **19.** $\log(e^x+e^{-x})+C$

20. $\frac{1}{2}\log(e^{2x}+e^{-2x})+C$ **21.** $\frac{1}{2}\tan(2x-3)-x+C$

22. $-\frac{1}{4}\tan(7-4x)+C$ **23.** $\frac{1}{2}(\sin^{-1}x)^2+C$

24. $\frac{1}{2}\log|2\sin x+3\cos x|+C$ **25.** $\frac{1}{(1-\tan x)}+C$

26. $2\sin\sqrt{x}+C$ **27.** $\frac{1}{3}(\sin 2x)^{\frac{3}{2}}+C$ **28.** $2\sqrt{1+\sin x}+C$

29. $\frac{1}{2}(\log\sin x)^2+C$ **30.** $-\log|1+\cos x|+C$ **31.** $\frac{1}{1+\cos x}+C$

32. $\frac{x}{2}-\frac{1}{2}\log|\cos x+\sin x|+C$ **33.** $\frac{x}{2}-\frac{1}{2}\log|\cos x-\sin x|+C$

34. $2\sqrt{\tan x}+C$ **35.** $\frac{1}{3}(1+\log x)^3+C$ **36.** $\frac{1}{3}(x+\log x)^3+C$

37. $-\frac{1}{4}\cos(\tan^{-1}x^4)+C$ **38.** D

39. B

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EXERCISE 7.3

1. $\frac{x}{2}-\frac{1}{8}\sin(4x+10)+C$

2. $-\frac{1}{14}\cos 7x+\frac{1}{2}\cos x+C$

3. $\frac{1}{4}\left[\frac{1}{12}\sin 12x+x+\frac{1}{8}\sin 8x+\frac{1}{4}\sin 4x\right]+C$

4. $-\frac{1}{2}\cos(2x+1)+\frac{1}{6}\cos^3(2x+1)+C$

5. $\frac{1}{6}\cos^6 x-\frac{1}{4}\cos^4 x+C$

6. $\frac{1}{4}\left[\frac{1}{6}\cos 6x-\frac{1}{4}\cos 4x-\frac{1}{2}\cos 2x\right]+C$

7. $\frac{1}{2}\left[\frac{1}{4}\sin 4x-\frac{1}{12}\sin 12x\right]+C$

8. $2\tan\frac{x}{2}-x+C$

9. $x-\tan\frac{x}{2}+C$

10. $\frac{3x}{8}-\frac{1}{4}\sin 2x+\frac{1}{32}\sin 4x+C$

11. $\frac{3x}{8}+\frac{1}{8}\sin 4x+\frac{1}{64}\sin 8x+C$

12. $x-\sin x+C$

13. $2(\sin x+x\cos\alpha)+C$

14. $-\frac{1}{\cos x+\sin x}+C$

15. $\frac{1}{6}\sec^3 2x-\frac{1}{2}\sec 2x+C$

16. $\frac{1}{3}\tan^3 x-\tan x+x+C$

17. $\sec x-\operatorname{cosec} x+C$

18. $\tan x+C$

19. $\log|\tan x|+\frac{1}{2}\tan^2 x+C$

20. $\log|\cos x+\sin x|+C$

21. $\frac{\pi x}{2}-\frac{x^2}{2}+C$

22. $\frac{1}{\sin(a-b)}\log\left|\frac{\cos(x-a)}{\cos(x-b)}\right|+C$

23. A

24. B

EXERCISE 7.4

1. $\tan^{-1}x^3+C$

2. $\frac{1}{2}\log\left|2x+\sqrt{1+4x^2}\right|+C$

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3. $\log\left|\dfrac{1}{2-x+\sqrt{x^2-4x+5}}\right|+C$

4. $\dfrac{1}{5}\sin^{-1}\dfrac{5x}{3}+C$

5. $\dfrac{3}{2\sqrt{2}}\tan^{-1}\sqrt{2}\,x^2+C$

6. $\dfrac{1}{6}\log\left|\dfrac{1+x^3}{1-x^3}\right|+C$

7. $\sqrt{x^2-1}-\log\left|x+\sqrt{x^2-1}\right|+C$

8. $\dfrac{1}{3}\log\left|x^3+\sqrt{x^6+a^6}\right|+C$

9. $\log\left|\tan x+\sqrt{\tan^2 x+4}\right|+C$

10. $\log\left|x+1+\sqrt{x^2+2x+2}\right|+C$

11. $\dfrac{1}{6}\tan^{-1}\left(\dfrac{3x+1}{2}\right)+C$

12. $\sin^{-1}\left(\dfrac{x+3}{4}\right)+C$

13. $\log\left|x-\dfrac{3}{2}+\sqrt{x^2-3x+2}\right|+C$

14. $\sin^{-1}\left(\dfrac{2x-3}{\sqrt{41}}\right)+C$

15. $\log\left|x-\dfrac{a+b}{2}+\sqrt{(x-a)(x-b)}\right|+C$

16. $2\sqrt{2x^2+x-3}+C$

17. $\sqrt{x^2-1}+2\log\left|x+\sqrt{x^2-1}\right|+C$

18. $\dfrac{5}{6}\log\left|3x^2+2x+1\right|-\dfrac{11}{3\sqrt{2}}\tan^{-1}\left(\dfrac{3x+1}{\sqrt{2}}\right)+C$

19. $6\sqrt{x^2-9x+20}+34\log\left|x-\dfrac{9}{2}+\sqrt{x^2-9x+20}\right|+C$

20. $-\sqrt{4x-x^2}+4\sin^{-1}\left(\dfrac{x-2}{2}\right)+C$

21. $\sqrt{x^2+2x+3}+\log\left|x+1+\sqrt{x^2+2x+3}\right|+C$

22. $\dfrac{1}{2}\log\left|x^2-2x-5\right|+\dfrac{2}{\sqrt{6}}\log\left|\dfrac{x-1-\sqrt{6}}{x-1+\sqrt{6}}\right|+C$

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23. $5\sqrt{x^2+4x+10}-7\log\left|x+2+\sqrt{x^2+4x+10}\right|+C$

24. B **25.** B

EXERCISE 7.5

1. $\log\dfrac{(x+2)^2}{|x+1|}+C$ **2.** $\dfrac{1}{6}\log\left|\dfrac{x-3}{x+3}\right|+C$

3. $\log|x-1|-5\log|x-2|+4\log|x-3|+C$

4. $\dfrac{1}{2}\log|x-1|-2\log|x-2|+\dfrac{3}{2}\log|x-3|+C$

5. $4\log|x+2|-2\log|x+1|+C$ **6.** $\dfrac{x}{2}+\log|x|-\dfrac{3}{4}\log|1-2x|+C$

7. $\dfrac{1}{2}\log|x-1|-\dfrac{1}{4}\log(x^2+1)+\dfrac{1}{2}\tan^{-1}x+C$

8. $\dfrac{2}{9}\log\left|\dfrac{x-1}{x+2}\right|-\dfrac{1}{3(x-1)}+C$ **9.** $\dfrac{1}{2}\log\left|\dfrac{x+1}{x-1}\right|-\dfrac{4}{x-1}+C$

10. $\dfrac{5}{2}\log|x+1|-\dfrac{1}{10}\log|x-1|-\dfrac{12}{5}\log|2x+3|+C$

11. $\dfrac{5}{3}\log|x+1|-\dfrac{5}{2}\log|x+2|+\dfrac{5}{6}\log|x-2|+C$

12. $\dfrac{x^2}{2}+\dfrac{1}{2}\log|x+1|+\dfrac{3}{2}\log|x-1|+C$

13. $-\log|x-1|+\dfrac{1}{2}\log(1+x^2)+\tan^{-1}x+C$

14. $3\log|x+2|+\dfrac{7}{x+2}+C$ **15.** $\dfrac{1}{4}\log\left|\dfrac{x-1}{x+1}\right|-\dfrac{1}{2}\tan^{-1}x+C$

16. $\dfrac{1}{n}\log\left|\dfrac{x^n}{x^n+1}\right|+C$ **17.** $\log\left|\dfrac{2-\sin x}{1-\sin x}\right|+C$

18. $x+\dfrac{2}{\sqrt{3}}\tan^{-1}\dfrac{x}{\sqrt{3}}-3\tan^{-1}\dfrac{x}{2}+C$ **19.** $\dfrac{1}{2}\log\left(\dfrac{x^2+1}{x^2+3}\right)+C$

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20. $\frac{1}{4}\log\left|\frac{x^4-1}{x^4}\right|+C$

21. $\log\left(\frac{e^x-1}{e^x}\right)+C$

22. B

23. A

EXERCISE 7.6

1. $-x\cos x+\sin x+C$

2. $-\frac{x}{3}\cos 3x+\frac{1}{9}\sin 3x+C$

3. $e^x(x^2-2x+2)+C$

4. $\frac{x^2}{2}\log x-\frac{x^2}{4}+C$

5. $\frac{x^2}{2}\log 2x-\frac{x^2}{4}+C$

6. $\frac{x^3}{3}\log x-\frac{x^3}{9}+C$

7. $\frac{1}{4}(2x^2-1)\sin^{-1}x+\frac{x\sqrt{1-x^2}}{4}+C$

8. $\frac{x^2}{2}\tan^{-1}x-\frac{x}{2}+\frac{1}{2}\tan^{-1}x+C$

9. $(2x^2-1)\frac{\cos^{-1}x}{4}-\frac{x}{4}\sqrt{1-x^2}+C$

10. $\left(\sin^{-1}x\right)^2x+2\sqrt{1-x^2}\sin^{-1}x-2x+C$

11. $-\sqrt{1-x^2}\cos^{-1}x+x+C$

12. $x\tan x+\log|\cos x|+C$

13. $x\tan^{-1}x-\frac{1}{2}\log(1+x^2)+C$

14. $\frac{x^2}{2}(\log x)^2-\frac{x^2}{2}\log x+\frac{x^2}{4}+C$

15. $\left(\frac{x^3}{3}+x\right)\log x-\frac{x^3}{9}-x+C$

16. $e^x\sin x+C$

17. $\frac{e^x}{1+x}+C$

18. $e^x\tan\frac{x}{2}+C$

19. $\frac{e^x}{x}+C$

20. $\frac{e^x}{(x-1)^2}+C$

21. $\frac{e^{2x}}{5}(2\sin x-\cos x)+C$

22. $2x\tan^{-1}x-\log(1+x^2)+C$

23. A

24. B

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EXERCISE 7.7

1. $\frac{1}{2}x\sqrt{4-x^2}+2\sin^{-1}\frac{x}{2}+C$

2. $\frac{1}{4}\sin^{-1}2x+\frac{1}{2}x\sqrt{1-4x^2}+C$

3. $\frac{(x+2)}{2}\sqrt{x^2+4x+6}+\log\left|x+2+\sqrt{x^2+4x+6}\right|+C$

4. $\frac{(x+2)}{2}\sqrt{x^2+4x+1}-\frac{3}{2}\log\left|x+2+\sqrt{x^2+4x+1}\right|+C$

5. $\frac{5}{2}\sin^{-1}\left(\frac{x+2}{\sqrt{5}}\right)+\frac{x+2}{2}\sqrt{1-4x-x^2}+C$

6. $\frac{(x+2)}{2}\sqrt{x^2+4x-5}-\frac{9}{2}\log\left|x+2+\sqrt{x^2+4x-5}\right|+C$

7. $\frac{(2x-3)}{4}\sqrt{1+3x-x^2}+\frac{13}{8}\sin^{-1}\left(\frac{2x-3}{\sqrt{13}}\right)+C$

8. $\frac{2x+3}{4}\sqrt{x^2+3x}-\frac{9}{8}\log\left|x+\frac{3}{2}+\sqrt{x^2+3x}\right|+C$

9. $\frac{x}{6}\sqrt{x^2+9}+\frac{3}{2}\log\left|x+\sqrt{x^2+9}\right|+C$

10. A

11. D

EXERCISE 7.8

1. 2

2. $\log\frac{3}{2}$

3. $\frac{64}{3}$

4. $\frac{1}{2}$

5. 0

6. $e^4(e-1)$

7. $\frac{1}{2}\log 2$

8. $\log\left(\frac{\sqrt{2}-1}{2-\sqrt{3}}\right)$

9. $\frac{\pi}{2}$

10. $\frac{\pi}{4}$

11. $\frac{1}{2}\log\frac{3}{2}$

12. $\frac{\pi}{4}$

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13. $\frac{1}{2}\log 2$

14. $\frac{1}{5}\log 6+\frac{3}{\sqrt{5}}\tan^{-1}\sqrt{5}$

15. $\frac{1}{2}(e-1)$

16. $5-\frac{5}{2}\left(9\log\frac{5}{4}-\log\frac{3}{2}\right)$

17. $\frac{\pi^4}{1024}+\frac{\pi}{2}+2$

18. 0

19. $3\log 2+\frac{3\pi}{8}$

20. $1+\frac{4}{\pi}-\frac{2\sqrt{2}}{\pi}$

21. D

22. C

EXERCISE 7.9

1. $\frac{1}{2}\log 2$

2. $\frac{64}{231}$

3. $\frac{\pi}{2}-\log 2$

4. $\frac{16\sqrt{2}}{15}(\sqrt{2}+1)$

5. $\frac{\pi}{4}$

6. $\frac{1}{\sqrt{17}}\log\frac{21+5\sqrt{17}}{4}$

7. $\frac{\pi}{8}$

8. $\frac{e^2(e^2-2)}{4}$

9. D

10. B

EXERCISE 7.10

1. $\frac{\pi}{4}$

2. $\frac{\pi}{4}$

3. $\frac{\pi}{4}$

4. $\frac{\pi}{4}$

5. 29

6. 9

7. $\frac{1}{(n+1)(n+2)}$

8. $\frac{\pi}{8}\log 2$

9. $\frac{16\sqrt{2}}{15}$

10. $\frac{\pi}{2}\log\frac{1}{2}$

11. $\frac{\pi}{2}$

12. π

13. 0

14. 0

15. 0

16. $-\pi\log 2$

17. $\frac{a}{2}$

18. 5

20. C

21. C

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MISCELLANEOUS EXERCISE ON CHAPTER 7

1. $\frac{1}{2}\log\left|\frac{x^2}{1-x^2}\right|+C$

2. $\frac{2}{3(a-b)}\left[(x+a)^{\frac{3}{2}}-(x+b)^{\frac{3}{2}}\right]+C$

3. $-\frac{2}{a}\sqrt{\frac{(a-x)}{x}}+C$

4. $-\left(1+\frac{1}{x^4}\right)^{\frac{1}{4}}+C$

5. $2\sqrt{x}-3x^{\frac{1}{3}}+6x^{\frac{1}{6}}-6\log(1+x^{\frac{1}{6}})+C$

6. $-\frac{1}{2}\log|x+1|+\frac{1}{4}\log(x^2+9)+\frac{3}{2}\tan^{-1}\frac{x}{3}+C$

7. $\sin a\log|\sin(x-a)|+x\cos a+C$

8. $\frac{x^3}{3}+C$

9. $\sin^{-1}\left(\frac{\sin x}{2}\right)+C$

10. $-\frac{1}{2}\sin 2x+C$

11. $\frac{1}{\sin(a-b)}\log\left|\frac{\cos(x+b)}{\cos(x+a)}\right|+C$

12. $\frac{1}{4}\sin^{-1}(x^4)+C$

13. $\log\left(\frac{1+e^x}{2+e^x}\right)+C$

14. $\frac{1}{3}\tan^{-1}x-\frac{1}{6}\tan^{-1}\frac{x}{2}+C$

15. $-\frac{1}{4}\cos^4 x+C$

16. $\frac{1}{4}\log(x^4+1)+C$

17. $\frac{[f(ax+b)]^{n+1}}{a(n+1)}+C$

18. $\frac{-2}{\sin\alpha}\sqrt{\frac{\sin(x+\alpha)}{\sin x}}+C$

19. $-2\sqrt{1-x}+\cos^{-1}\sqrt{x}+\sqrt{x-x^2}+C$

20. $e^x\tan x+C$

21. $-2\log|x+1|-\frac{1}{x+1}+3\log|x+2|+C$

22. $\frac{1}{2}\left[x\cos^{-1}x-\sqrt{1-x^2}\right]+C$

23. $-\frac{1}{3}\left(1+\frac{1}{x^2}\right)^{\frac{3}{2}}\left[\log\left(1+\frac{1}{x^2}\right)-\frac{2}{3}\right]+C$

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24. $e^{\frac{\pi}{2}}$

25. $\frac{\pi}{8}$

26. $\frac{\pi}{6}$

27. $2\sin^{-1}\frac{(\sqrt{3}-1)}{2}$

28. $\frac{4\sqrt{2}}{3}$

29. $\frac{1}{40}\log 9$

30. $\frac{\pi}{2}-1$

31. $\frac{19}{2}$

38. A

39. B

40. D

EXERCISE 8.1

1. 12π **2.** 6π **3.** A **4.** B

Miscellaneous Exercise on Chapter 8

1. (i) $\frac{7}{3}$ (ii) 624.8

2. 9 **3.** 4 **4.** D **5.** C

EXERCISE 9.1

1. Order 4; Degree not defined

2. Order 1; Degree 1

3. Order 2; Degree 1

4. Order 2; Degree not defined

5. Order 2; Degree 1

6. Order 3; Degree 2

7. Order 3; Degree 1

8. Order 1; Degree 1

9. Order 2; Degree 1

10. Order 2; Degree 1

11. D

12. A

EXERCISE 9.2

11. D

12. D

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EXERCISE 9.3

1. $y = 2\tan\frac{x}{2} - x + C$
2. $y = 2\sin(x + C)$
3. $y = 1 + Ae^{-x}$
4. $\tan x \tan y = C$
5. $y = \log(e^x + e^{-x}) + C$
6. $\tan^{-1} y = x + \frac{x^3}{3} + C$
7. $y = e^{cx}$
8. $x^{-4} + y^{-4} = C$
9. $y = x\sin^{-1}x + \sqrt{1-x^2} + C$
10. $\tan y = C(1 - e^x)$
11. $y = \frac{1}{4}\log\left[(x+1)^2(x^2+1)^3\right] - \frac{1}{2}\tan^{-1}x + 1$
12. $y = \frac{1}{2}\log\left(\frac{x^2-1}{x^2}\right) - \frac{1}{2}\log\frac{3}{4}$
13. $\cos\left(\frac{y-2}{x}\right) = a$
14. $y = \sec x$
15. $2y - 1 = e^x(\sin x - \cos x)$
16. $y - x + 2 = \log(x^2(y+2)^2)$
17. $y^2 - x^2 = 4$
18. $(x+4)^2 = y + 3$
19. $(63t + 27)^{\frac{1}{3}}$
20. 6.93%
21. Rs 1648
22. $\frac{2\log 2}{\log\left(\frac{11}{10}\right)}$
23. A

EXERCISE 9.4

1. $(x-y)^2 = Cx\, e^{\frac{-y}{x}}$
2. $y = x\log|x| + Cx$
3. $\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log(x^2+y^2) + C$
4. $x^2 + y^2 = Cx$
5. $\frac{1}{2\sqrt{2}}\log\left|\frac{x+\sqrt{2}y}{x-\sqrt{2}y}\right| = \log|x| + C$
6. $y + \sqrt{x^2+y^2} = Cx^2$
7. $xy\cos\left|\frac{y}{x}\right| = C$
8. $x\left[1-\cos\left(\frac{y}{x}\right)\right] = C\sin\left(\frac{y}{x}\right)$

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9. $cy = \log\left|\frac{y}{x}\right| - 1$

10. $ye^{\frac{x}{y}} + x = C$

11. $\log(x^2 + y^2) + 2\tan^{-1}\frac{y}{x} = \frac{\pi}{2} + \log 2$

12. $y + 2x = 3x^2 y$

13. $\cot\left(\frac{y}{x}\right) = \log|ex|$

14. $\cos\left(\frac{y}{x}\right) = \log|ex|$

15. $y = \frac{2x}{1 - \log|x|}\ (x \neq 0,\ x \neq e)$

16. C

17. D

EXERCISE 9.5

1. $y = \frac{1}{5}(2\sin x - \cos x) + C\,e^{-2x}$

2. $y = e^{-2x} + Ce^{-3x}$

3. $xy = \frac{x^4}{4} + C$

4. $y(\sec x + \tan x) = \sec x + \tan x - x + C$

5. $y = (\tan x - 1) + Ce^{-\tan x}$

6. $y = \frac{x^2}{16}(4\log|x| - 1) + Cx^{-2}$

7. $y\log x = \frac{-2}{x}(1 + \log|x|) + C$

8. $y = (1+x)^{-1}\log|\sin x| + C(1+x^2)^{-1}$

9. $y = \frac{1}{x} - \cot x + \frac{C}{x\sin x}$

10. $(x + y + 1) = C\,e^y$

11. $x = \frac{y^2}{3} + \frac{C}{y}$

12. $x = 3y^2 + Cy$

13. $y = \cos x - 2\cos^2 x$

14. $y(1 + x^2) = \tan^{-1} x - \frac{\pi}{4}$

15. $y = 4\sin^3 x - 2\sin^2 x$

16. $x + y + 1 = e^x$

17. $y = 4 - x - 2e^x$

18. C

19. D

Miscellaneous Exercise on Chapter 9

1. (i) Order 2; Degree 1 (ii) Order 1; Degree 3

(iii) Order 4; Degree not defined

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4. $\sin^{-1}y + \sin^{-1}x = C$

6. $\cos y = \dfrac{\sec x}{\sqrt{2}}$

7. $\tan^{-1} y + \tan^{-1}(e^x) = \dfrac{\pi}{2}$

8. $e^{\frac{x}{y}} = y + C$

9. $\log |x - y| = x + y + 1$

10. $y e^{2\sqrt{x}} = (2\sqrt{x} + C)$

11. $y \sin x = 2x^2 - \dfrac{\pi^2}{2}$ $(\sin x \neq 0)$

12. $y = \log\left|\dfrac{2x+1}{x+1}\right|, x \neq -1$

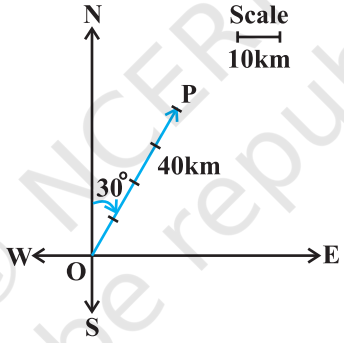
13. C

14. C

15. C

EXERCISE 10.1

1. In the adjoining figure, the vector $\overrightarrow{OP}$ represents the required displacement.



2. (i) scalar (ii) vector (iii) scalar (iv) scalar (v) scalar (vi) vector

3. (i) scalar (ii) scalar (iii) vector (iv) vector (v) scalar

4. (i) Vectors $\vec{a}$ and $\vec{b}$ are coinitial

(ii) Vectors $\vec{b}$ and $\vec{d}$ are equal

(iii) Vectors $\vec{a}$ and $\vec{c}$ are collinear but not equal

5. (i) True (ii) False (iii) False (iv) False

EXERCISE 10.2

1. $|\vec{a}| = \sqrt{3}, |\vec{b}| = \sqrt{62}, |\vec{c}| = 1$

2. An infinite number of possible answers.

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3. An infinite number of possible answers.

4. $x = 2, y = 3$

5. -7 and 6; $-7\hat{i}$ and $6\hat{j}$

6. $-4\hat{j}-\hat{k}$

7. $\frac{1}{\sqrt{6}}\hat{i}+\frac{1}{\sqrt{6}}\hat{j}+\frac{2}{\sqrt{6}}\hat{k}$

8. $\frac{1}{\sqrt{3}}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}+\frac{1}{\sqrt{3}}\hat{k}$

9. $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}$

10. $\frac{40}{\sqrt{30}}\hat{i}-\frac{8}{\sqrt{30}}\hat{j}+\frac{16}{\sqrt{30}}\hat{k}$

12. $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$

13. $-\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}$

15. (i) $-\frac{1}{3}\hat{i}+\frac{4}{3}\hat{j}+\frac{1}{3}\hat{k}$ (ii) $-3\hat{i}+3\hat{k}$

16. $3\hat{i}+2\hat{j}+\hat{k}$

18. (C)

19. (B), (C), (D)

EXERCISE 10.3

1. $\frac{\pi}{4}$

2. $\cos^{-1}\left(\frac{5}{7}\right)$

3. 0

4. $\frac{60}{\sqrt{114}}$

6. $\frac{16\sqrt{2}}{3\sqrt{7}}, \frac{2\sqrt{2}}{3\sqrt{7}}$

7. $6|\vec{a}|^2+11\vec{a}.\vec{b}-35|\vec{b}|^2$

8. $|\vec{a}|=1, |\vec{b}|=1$

9. $\sqrt{13}$

10. 8

12. Vector $\vec{b}$ can be any vector

13. $\frac{-3}{2}$

14. Take any two non-zero perpendicular vectors $\vec{a}$ and $\vec{b}$

15. $\cos^{-1}\left(\frac{10}{\sqrt{102}}\right)$

18. (D)

EXERCISE 10.4

1. $19\sqrt{2}$

2. $\pm\frac{2}{3}\hat{i} \quad \frac{2}{3}\hat{j} \quad \frac{1}{3}\hat{k}$

3. $\frac{\pi}{3}; \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}$

5. $3, \frac{27}{2}$

6. Either $|\vec{a}|=0$ or $|\vec{b}|=0$

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8. No; take any two nonzero collinear vectors

9. $\frac{\sqrt{61}}{2}$ **10.** $15\sqrt{2}$ **11.** (B) **12.** (C)

Miscellaneous Exercise on Chapter 10

1. $\frac{\sqrt{3}}{2}\hat{i}+\frac{1}{2}\hat{j}$

2. $x_2-x_1, y_2-y_1, z_2-z_1; \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

3. $\frac{-5}{2}\hat{i}+\frac{3\sqrt{3}}{2}\hat{j}$

4. No; take $\vec{a}$, $\vec{b}$ and $\vec{c}$ to represent the sides of a triangle.

5. $\pm\frac{1}{\sqrt{3}}$ **6.** $\frac{3}{2}\sqrt{10}\,\hat{i}+\frac{\sqrt{10}}{2}\hat{j}$ **7.** $\frac{3}{\sqrt{22}}\hat{i}-\frac{3}{\sqrt{22}}\hat{j}+\frac{2}{\sqrt{22}}\hat{k}$

8. 2 : 3 **9.** $3\vec{a}+5\vec{b}$ **10.** $\frac{1}{7}(3\hat{i}-6\hat{j}+2\hat{k})$; $11\sqrt{5}$

12. $\frac{1}{3}(160\hat{i}-5\hat{j}+70\hat{k})$ **13.** $\lambda = 1$ **16.** (B)

17. (D) **18.** (C) **19.** (B)

EXERCISE 11.1

1. $0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$ **2.** $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$ **3.** $\frac{-9}{11}, \frac{6}{11}, \frac{-2}{11}$

5. $\frac{-2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}, \frac{3}{17}$; $\frac{-2}{\sqrt{17}}, \frac{-3}{\sqrt{17}}, \frac{-2}{\sqrt{17}}$; $\frac{4}{\sqrt{42}}, \frac{5}{\sqrt{42}}, \frac{-1}{\sqrt{42}}$

EXERCISE 11.2

4. $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$, where λ is a real number

5. $\vec{r} = 2\hat{i} - \hat{j} + 4\hat{k} + \lambda(\hat{i} + 2\hat{j} - \hat{k})$ and cartesian form is

$$\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-4}{-1}$$

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6. $\dfrac{x+2}{3} = \dfrac{y-4}{5} = \dfrac{z+5}{6}$

7. $\vec{r} = (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} + 2\hat{k})$

8. (i) $\theta = \cos^{-1}\left(\dfrac{19}{21}\right)$ (ii) $\theta = \cos^{-1}\left(\dfrac{8}{5\sqrt{3}}\right)$

9. (i) $\theta = \cos^{-1}\left(\dfrac{26}{9\sqrt{38}}\right)$ (ii) $\theta = \cos^{-1}\left(\dfrac{2}{3}\right)$

10. $p = \dfrac{70}{11}$ **12.** $\dfrac{3\sqrt{2}}{2}$ **13.** $2\sqrt{29}$

14. $\dfrac{3}{\sqrt{19}}$ **15.** $\dfrac{8}{\sqrt{29}}$

Miscellaneous Exercise on Chapter 11

1. 90° **2.** $\dfrac{x}{1} = \dfrac{y}{0} = \dfrac{z}{0}$ **3.** $k = \dfrac{-10}{7}$

4. 9 **5.** $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

EXERCISE 12.1

1. Maximum Z = 16 at (0, 4)

2. Minimum Z = – 12 at (4, 0)

3. Maximum Z = $\dfrac{235}{19}$ at $\left(\dfrac{20}{19}, \dfrac{45}{19}\right)$

4. Minimum Z = 7 at $\left(\dfrac{3}{2}, \dfrac{1}{2}\right)$

5. Maximum Z = 18 at (4, 3)

6. Minimum Z = 6 at all the points on the line segment joining the points (6, 0) and (0, 3).

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7. Minimum Z = 300 at (60, 0);

Maximum Z = 600 at all the points on the line segment joining the points (120, 0) and (60, 30).

8. Minimum Z = 100 at all the points on the line segment joining the points (0, 50) and (20, 40);

Maximum Z = 400 at (0, 200)

9. Z has no maximum value

10. No feasible region, hence no maximum value of Z.

EXERCISE 13.1

1. $P(E|F) = \frac{2}{3}, P(F|E) = \frac{1}{3}$ **2.** $P(A|B) = \frac{16}{25}$

3. (i) 0.32 (ii) 0.64 (iii) 0.98

4. $\frac{11}{26}$

5. (i) $\frac{4}{11}$ (ii) $\frac{4}{5}$ (iii) $\frac{2}{3}$

6. (i) $\frac{1}{2}$ (ii) $\frac{3}{7}$ (iii) $\frac{6}{7}$

7. (i) 1 (ii) 0

8. $\frac{1}{6}$ **9.** 1 **10.** (a) $\frac{1}{3}$, (b) $\frac{1}{9}$

11. (i) $\frac{1}{2}, \frac{1}{3}$ (ii) $\frac{1}{2}, \frac{2}{3}$ (iii) $\frac{3}{4}, \frac{1}{4}$

12. (i) $\frac{1}{2}$ (ii) $\frac{1}{3}$ **13.** $\frac{5}{9}$

14. $\frac{1}{15}$ **15.** 0 **16.** C **17.** D

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EXERCISE 13.2

1. $\frac{3}{25}$ **2.** $\frac{25}{102}$ **3.** $\frac{44}{91}$

4. A and B are independent **5.** A and B are not independent

6. E and F are not independent

7. (i) $p = \frac{1}{10}$ (ii) $p = \frac{1}{5}$

8. (i) 0.12 (ii) 0.58 (iii) 0.3 (iv) 0.4

9. $\frac{3}{8}$ **10.** A and B are not independent

11. (i) 0.18 (ii) 0.12 (iii) 0.72 (iv) 0.28

12. $\frac{7}{8}$ **13.** (i) $\frac{16}{81}$, (ii) $\frac{20}{81}$, (iii) $\frac{40}{81}$

14. (i) $\frac{2}{3}$, (ii) $\frac{1}{2}$ **15.** (i), (ii) **16.** (a) $\frac{1}{5}$, (b) $\frac{1}{3}$, (c) $\frac{1}{2}$

17. D **18.** B

EXERCISE 13.3

1. $\frac{1}{2}$ **2.** $\frac{2}{3}$ **3.** $\frac{9}{13}$ **4.** $\frac{12}{13}$

5. $\frac{22}{133}$ **6.** $\frac{4}{9}$ **7.** $\frac{1}{52}$ **8.** $\frac{1}{4}$

9. $\frac{2}{9}$ **10.** $\frac{8}{11}$ **11.** $\frac{5}{34}$ **12.** $\frac{11}{50}$

13. A **14.** C

Miscellaneous Exercise on Chapter 13

1. (i) 1 (ii) 0

2. (i) $\frac{1}{3}$ (ii) $\frac{1}{2}$

3. $\frac{20}{21}$

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4. $1-\sum_{r=7}^{10} {}^{10}C_r (0.9)^r (0.1)^{10-r}$

5. $\frac{2}{7}$

6. $\frac{1}{15}, \frac{2}{5}, \frac{8}{15}$

7. $\frac{14}{29}$

8. $\frac{3}{16}$

9. (i) 0.5 (ii) 0.05

10. $\frac{16}{31}$

11. A

12. C

13. B

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Notes

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